

C&E BOOKSHOP BOOK SALES RECORDING SYSTEM

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ABSTRACT

This project is a desktop-based Book Sales Recording System for **C&E Bookshop**. It helps bookstore staff record book sales, calculate total prices and change, manage book information, and monitor book stocks. It also displays stock status, such as In Stock, Low Stock, and Out of Stock. The system includes a dashboard for viewing sales information, a book catalog with archiving, sales history, and account management features.

The system is developed using **Java GUI in NetBeans** and **MySQL** for storing data. **phpMyAdmin** is used to manage the database during development.

The project follows an **Agile SDLC with Scrum**. Development is organized into small stages to allow improvements and corrections during the process. The documentation also covers system requirements, feasibility, risks, schedule, budget, and project activities.

1. PROJECT BACKGROUND

1.1 Client / Organization Background

C&E Bookshop is a bookstore that sells books and educational materials to customers. The system is designed for the cashier, inventory staff, and store manager or owner. It will be a Java-based desktop application installed on the bookstore's computer to help manage book sales and inventory.

1.2 Current Process

In a manual bookstore process, customers choose their books and bring them to the cashier. The cashier checks the book details, calculates the total price and change, and records the sale in a logbook or spreadsheet. Book stocks are also counted and updated manually to identify which books are running low or are out of stock. The proposed system aims to improve this process by recording sales digitally, calculating transaction amounts, and updating book quantities through the database.

1.3 Identified Problems

Based on the manual process, the following problems may occur:

- Manual sales recording and computation can lead to calculation errors during peak checkout times.

- Tracking stock levels manually makes it hard to identify low-stock or out-of-stock books quickly.

- Paper-based records or unorganized sales logs can easily be damaged, lost, or misplaced.

- Reviewing past transactions and daily revenue summaries takes extra time and effort.

- Unsecured user access can make sales logs and inventory records vulnerable to unauthorized changes.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of this project is to develop a simple desktop-based book sales recording and inventory system for C&E Bookshop. It aims to help bookstore staff record sales, calculate transaction totals and change, manage book information, monitor stock levels, and reduce unnecessary manual work during daily store operations.

2.2 Main Project Objective

To design and develop a desktop-based book sales recording system for C&E Bookshop that supports faster sales transactions and organized inventory tracking.

2.3 Specific Objectives

- To allow cashiers to select books, enter quantities, and record sales through a digital interface.

- To automatically calculate the total price and change based on the transaction amount and payment received.

- To manage book details, prices, categories, and stock quantities.

- To display stock status automatically as In Stock, Low Stock, or Out of Stock.

- To provide authorized users with login access based on their assigned roles, such as Admin and Cashier.

- To provide book archiving so that books can be removed from the active catalog without necessarily deleting their records.

- To provide account settings for users to change their own passwords.

- To display sales history and summary information to help authorized staff monitor bookstore transactions.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

The proposed desktop application will include the following major functions:

User/Area	Included Functions
Cashier	Login, search books by title/ISBN, process sales, calculate total price and change, select payment method (Cash/GCash), and view recent sales activity.
Inventory / Admin	Manage the book catalog (add, edit, archive books), monitor stock levels (In Stock, Low Stock, Out of Stock), manage system users and roles, and access account settings and password recovery features.
Manager / Authorized User	View total sales summaries, overall books sold, low stock alerts, and track historical transaction records.
System	Store sales and inventory records locally in a MySQL database and maintain user security controls.

3.2 Project Limitations

- The first version will focus on local, in-store sales transactions and will not support online book ordering or e-commerce.

- The project will not include advanced accounting, profit-margin analysis, or employee payroll features.

- It does not connect directly to card payment terminals or online payment gateways.

- The system will depend on a single desktop computer connected to a local MySQL database setup.

- Barcode scanner integration is not included in the first version and may be added in the future.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically feasible for a student development team because it can be created using Java GUI (NetBeans Swing) for the desktop application, MySQL for database management, and phpMyAdmin for database maintenance. These tools can run on a standard development computer and are suitable for developing essential sales recording, transaction, and inventory management functions.

4.2 Operational Feasibility

The system is practical because it follows the bookstore's normal sales process. It is designed to be simple and easy for staff to use, with clear buttons, readable fields, and a sales table for checking transaction details before recording a sale.

4.3 Economic Feasibility

The project is economically feasible because it can use free tools such as NetBeans and XAMPP. Possible costs include transportation, printing, storage, and other deployment expenses. The system can help reduce calculation errors and make inventory management easier, which can provide benefits to the bookstore.

4.4 Project Charter Summary

Charter Item	Sample Entry
Project Title	Desktop-Based Book Sales Recording System for C&E Bookshop
Project Purpose	Improve the bookstore's sales recording and stock monitoring process through a desktop application.
Primary Users	Cashier, inventory staff, and store manager or owner.
Major Deliverable	A desktop-based book sales recording system with book management, sales recording, inventory monitoring, dashboard summaries, account management, and project documentation with a user guide.
Main Constraint	Limited development time, available computer hardware, and student-level resources.

Project Success	Core sales and inventory functions work correctly and store staff can record transactions smoothly using the system.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Bookstore Owner / Manager	Approves project direction, reviews sales metrics, and monitors stock levels.
Cashier	Main user who processes book sales, collects payments, and computes change.
Inventory Staff	Updates book catalog details, adds new stock, and checks low-stock status.
Customers	Indirect users affected by fast, accurate checkout counter service.
Development Team	Analyzes, designs, develops, tests, and documents the proposed system.

5.2 Requirements Gathering Methods

The development team may use the following methods to collect project requirements:

- Client interview to understand the current book sales and inventory monitoring process.

- Observation of how book sales transactions, payments, and stock checks are performed.

- Review of existing sales receipts, book logs, and inventory inventory records.

- Follow-up consultation to confirm unclear software features and database requirements.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow users to log in based on their assigned role (Admin or Cashier) and restrict access to functions according to their permissions.
FR-02	The system shall display available books, ISBN numbers, categories, and prices.
FR-03	The system shall allow the cashier to select books, adjust quantities, and calculate total price and change.

FR-04	The system shall record transaction details including payment amount and payment method (Cash/GCash).
FR-05	The system shall save completed sales records and update inventory quantity automatically.
FR-06	The system shall allow authorized users to view recent sales activity and total revenue metrics.
FR-07	The system shall allow only authorized Admin users to add, edit, and archive books in the catalog. Cashiers shall not have access to these functions.
FR-08	The system shall categorize and display inventory status (In Stock, Low Stock, Out of Stock).
FR-09	The system shall allow users to change their passwords through Account Settings after verifying their current password.
FR-10	The system shall provide an account recovery feature for the administrator using a recovery code.
FR-11	The system shall allow authorized users to search sales records and filter transactions by date.

5.4 Non-Functional Requirements

Category	Sample Requirement
Usability	The sales recording interface should be clear and simple enough for cashiers to operate with minimal steps.
Performance	Common actions such as loading the catalog or recording a sale should respond quickly on the counter computer.
Security	The system shall restrict book management and user management functions to authorized Admin users. Cashiers shall only access the functions permitted for their role.
Reliability	Recorded sales and inventory changes must save correctly to MySQL and remain intact after closing the app.
Maintainability	Book categories, stock counts, and prices should be updated in the database without modifying source code.

5.5 Sample User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As a cashier, I want to select books and enter quantities so that I can process customer purchases quickly.	Selected book title and price populate, quantity can be entered, and total price with change calculates automatically.
As a cashier, I want to select payment methods (Cash/GCash) so that transactions are recorded accurately.	Payment method drop-down allows selection, payment amount is validated against total price, and transaction saves.
As a manager, I want to monitor inventory status and low stock items so that I know when to reorder books.	Dashboard displays count of low stock items, and inventory list flags books with status labels (In Stock/Low Stock/Out of Stock).

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The project uses an Agile development methodology with the Scrum framework. The work is divided into short development cycles called sprints. In each sprint, the team chooses important tasks, develops and tests features, reviews the results, and makes improvements based on feedback. This helps the team track progress, handle changes, and focus on the important needs of C&E Bookshop.

6.2 SDLC Model Used

The project uses the Agile SDLC model, where the system is developed gradually through repeated iterations. Each iteration includes planning, requirements analysis, design**,** development, testing, and review. Feedback from the client or instructor is used to improve the system in the next iteration. This allows the team to make changes and improvements during development.

6.3 SDLC Model Figure

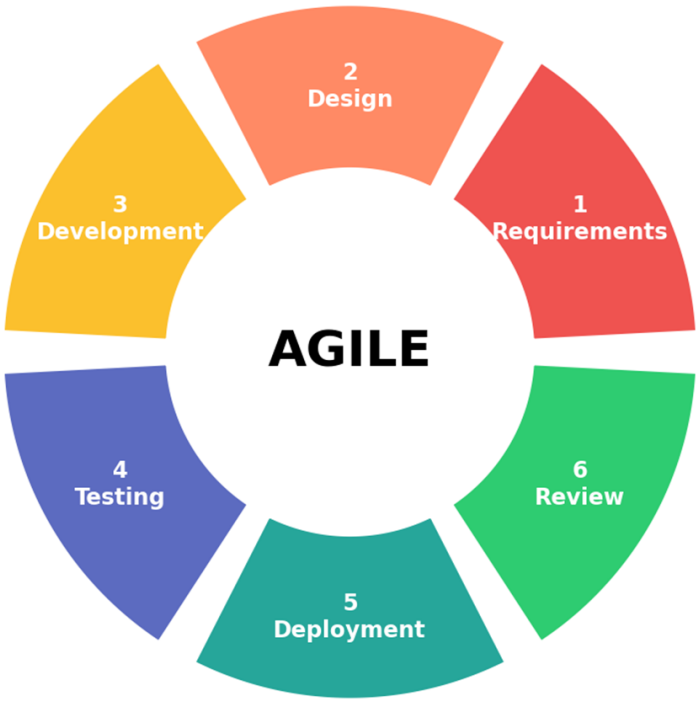


Figure 1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and technologies are planned for the development of the desktop-based sales recording system:

Software / Technology	Use in the Project

Java GUI (NetBeans Swing)	Used to create the desktop user interface and implement main sales, inventory, and login screens.
MySQL	Used as the database management system to store books, categories, users, and sales transaction logs.
phpMyAdmin	Used by the team to design, view, organize, and manage MySQL database tables and sample records.

Java GUI will serve as the application layer that bookstore staff interact with, while MySQL will store the project data. phpMyAdmin will mainly support database setup and administration during development and maintenance.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile SDLC with Scrum is suitable for this project because the bookstore's needs may become clearer as users see and use the system. The use of short sprints allows the team to develop important features first, show progress, receive feedback, and fix problems before continuing to the next tasks. It also helps the team stay organized and manage the limited project time.

6.6 Expected Development Approach

- Prepare and prioritize the initial system requirements.

- Divide the work into small features or backlog items.

- Develop a working set of features during each sprint.

- Test completed features before adding more work.

- Present progress to the client or instructor for feedback.

- Improve the system based on accepted feedback.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to the Book Sales Recording System

Planning	Identify the book sales problems, project objectives, user roles (Admin/Cashier), scope, and constraints.
Requirements Analysis	Identify the functional and non-functional requirements for book management, sales recording, inventory monitoring, account management, and user access permissions.
System Design	Design the Login, Dashboard, Books, Add New Book, Sales, Inventory, Account Settings, and Admin Account Recovery interfaces, along with the navigation flow and MySQL database structure.
Development	Develop the login system, dashboard, book catalog management, book archiving, sales recording, payment and change calculations, automatic stock updates, and role-based access controls.
Testing	Test login permissions, book management, book searches, sales calculations, payment validation, inventory updates, stock status indicators, and account management functions.
Deployment	Prepare the Java desktop application and local MySQL database for installation on the target bookstore computer.
Maintenance	Fix identified errors, update book records, maintain database backups, and improve the system when necessary.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The team must identify potential events that could impact the project's schedule, cost, scope, or quality, ensuring that risks are realistic and linked to the chosen project.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
Client changes important requirements during development.	Medium	High	Confirm requirements per sprint and record approved changes before development continues.
Development takes longer than planned.	Medium	High	Prioritize core features, monitor weekly progress, and adjust low-priority tasks.
Project files or database are lost.	Low	High	Use source control and maintain regular local and cloud backups.
A team member becomes unavailable.	Medium	Medium	Document work, share project access, and assign backup responsibilities.
Target computer has limited hardware capability.	Medium	Medium	Test the application on hardware close to the client's actual computer before deployment.
Users find the book sales recording flow confusing.	Medium	Medium	Perform early user testing and simplify screens based on feedback.
Cashiers gain unauthorized access to administrative functions.	Medium	High	Test role-based access controls to ensure that only Admin users can add, edit, and archive books.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule should show the order of major tasks, expected duration, dependencies, and important milestones. The schedule below is only a sample and should be adjusted to the actual academic timeline.

9.2 Sample Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Client Study	1 week	Week 1	Problem, objectives, scope
Requirements Gathering	1 week	Week 2	Confirmed requirements
System and UI Design	2 weeks	Weeks 3–4	Wireframes and database design
Sprint 1 Development	2 weeks	Weeks 5–6	Login, catalog display, basic sales entry
Sprint 2 Development	2 weeks	Weeks 7–8	Totals, payment/change calculation, saving transactions, sales history
Sprint 3 Development	2 weeks	Weeks 9–10	Inventory management, stock status indicators, book archiving, account settings, and role-based access controls.
Testing and Revision	1 week	Week 11	Corrected release candidate
Deployment and Final Documentation	1 week	Week 12	Installed system and final documents

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 2
Initial design approved	End of Week 4
Core sales recording functions working	End of Week 8
Feature-complete version	End of Week 10

Testing completed	End of Week 11
Final deployment / presentation	End of Week 12

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The project budget summarizes the financial resources needed to build and deploy the application for C&E Bookshop. Open-source development software is utilized to keep overall project expenses low.

Cost Item	Basis	Estimated Cost
Development Software	Free open-source tools (NetBeans IDE, Java JDK)	PHP 0
Database Software	Free open-source software (MySQL, phpMyAdmin via XAMPP)	PHP 0
Transportation	Travel costs for bookstore visits and client feedback meetings	PHP 1,500
Printing / Documentation	User manuals, system documentation, and review forms	PHP 600
Backup Storage / USB	Dedicated flash drive for offline database and project code backups	PHP 500
Hardware Adjustment	Budget allowance for peripheral setup or testing hardware	PHP 1,500
Contingency	Emergency funds for unexpected project needs	PHP 300
TOTAL ESTIMATED COST		PHP 4,400

10.2 Cost Justification

The budget is limited to operational, travel, and documentation costs required for client collaboration and deployment. Development labor is covered by the student team at no cost. Free development tools such as NetBeans and XAMPP keep software expenses at zero, while funds for transportation, hardware adjustments, and backups protect project security and smooth deployment at C&E Bookshop.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Sample Responsibility
Product Owner	Represents C&E Bookshop requirements, prioritizes software features, and validates completed sales/inventory functions
Scrum Master	Facilitates team meetings, resolves technical blockers, and keeps development on schedule .
Development Team	Programs the Java GUI, configures MySQL database tables, and executes system testing.

11.2 Sample Product Backlog

Priority	Backlog Item	Expected Result
High	User Login Authentication	Admin and Cashier log in and access functions according to their assigned roles.

High	Book Catalog Display	Displays book titles, ISBN numbers, prices, categories, and stock quantities.
High	Sales Entry & Cart	Allows cashiers to select books, enter quantities, and calculate transaction totals.
High	Automatic Computation	Calculates the total price and change based on the payment received.
High	Transaction Recording	Saves completed sales to MySQL and updates stock quantities.
Medium	Book Catalog Management	Allows Admin users to add, edit, and archive books.
Medium	Stock Status Monitoring	Displays In Stock, Low Stock, and Out of Stock indicators.
Medium	Transaction History Log	Allows authorized users to review sales records and available sales summaries.
Low	Report Exporting	Optional future feature if exporting reports is not included in the current system.

11.3 Sample Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Build initial desktop navigation and catalog screens.	Create the login window, database connection, book search, and catalog display.

Sprint 2	Implement sales calculation and database saving.	Develop book selection, quantity entry, total price and change calculations, payment method selection, and sales recording.
Sprint 3	Build inventory and stock monitoring tools.	Implement automatic stock updates, stock status indicators, book management and archiving for Admin users, dashboard summaries, and account settings.
Sprint 4	System verification and deployment readiness.	Test login permissions, sales calculations, inventory updates, book archiving, account features, and resolve identified errors before deployment.

11.4 Scrum Events

Scrum Event	How the Team Will Apply It
Sprint Planning	Select target user stories from the backlog and establish sprint deliverables.
Daily Scrum / Team Check-in	Team updates should be held regularly to monitor coding progress and pinpoint technical issues.
Sprint Review	Present the completed features to the bookstore manager or instructor for feedback.
Sprint Retrospective	Evaluate and address technical bottlenecks to enhance programming workflow for the upcoming sprint.

11.5 Simple Scrum Progress Tracking

The team uses a Kanban-style board to track development tasks by placing them under To Do, In Progress, and Completed. Tasks are assigned to team members and checked regularly to monitor the progress of the project. This helps the team keep track of completed features, remaining tasks, and the overall development of the C&E Bookshop system.

12. PROJECT SUMMARY

The C&E Bookshop Book Sales Recording System is designed to enhance store operations by replacing manual sales calculations and recordkeeping with a desktop application. This system enables users to record sales, calculate totals, manage payment methods, and monitor inventory. Key features include book management, archiving, dashboard summaries, and sales history. Developed using Java GUI in NetBeans and MySQL, it allows for distinct access for Admin and Cashier users, with administrative functions limited to authorized users. The project follows the Agile SDLC model, utilizing Scrum for development and testing, and focuses on in-store sales with a local MySQL database.